

Kozma–Nitzan Question 8: exact statement, boundary counterexample, and a formalization-ready core

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Abstract

The printed Question 8 in Kozma–Nitzan minimizes $\mathbb{P}(a \leftrightarrow b, o \nleftrightarrow A)$, not $\mathbb{P}(o \leftrightarrow A, a \leftrightarrow b)$. This distinction is essential. Under the usual reading that the desired inequality must hold for every chosen minimizer, the printed question is false when edge probabilities in $[0, 1]$ are allowed: an exact four-vertex example is given below. The unique-minimizer and positive-avoidance versions are not refuted by that example and remain open here.

This note gives a statement-faithful formulation, the complete counterexample, several proved reductions, and an atomized proof interface for Lean. In particular, the global unique-minimizer version is equivalent to the version for every minimizer under the hypothesis $\mathbb{P}(o \nleftrightarrow A) > 0$. The reduction uses a support-graph lemma for a zero score and a pendant-relay tie breaker for a positive score. No open claim is used as a proved lemma.

1 Model and exact notation

Let $G = (V, E)$ be a finite graph. Every edge $e \in E$ is independently open with probability $p_e \in [0, 1]$. Connectivity in the random open subgraph is denoted by $x \leftrightarrow y$, and connectivity to a nonempty finite set $A \subseteq V$ by $x \leftrightarrow A$. Connectivity is reflexive. Fix vertices $o, b \in V$, and put

$$U := \{o \leftrightarrow A\}, \quad D := U^c = \{o \nleftrightarrow A\}. \quad (1)$$

For every $x \in A$, define

$$L_x := \mathbb{P}(U, x \leftrightarrow b), \quad \rho_x := \mathbb{P}(D, x \leftrightarrow b), \quad R := \mathbb{P}(U, o \leftrightarrow b). \quad (2)$$

The elementary partition $U \sqcup D$ gives

$$\mathbb{P}(x \leftrightarrow b) = L_x + \rho_x \quad (x \in A). \quad (3)$$

With this notation, display (41) in the paper is $L_a \leq R$, while the selection rule in its Question 8 is

$$a \in \arg \min_{x \in A} \rho_x = \arg \min_{x \in A} \mathbb{P}(x \leftrightarrow b, o \nleftrightarrow A). \quad (4)$$

The slash in $o \nleftrightarrow A$ is visible in the rendered original PDF. Text extraction from that PDF can display the glyph incorrectly as “ $o = A$ ”, which is the source of the earlier transcription error.

2 The formulations that must not be conflated

The phrase “let a be the point minimising” is ambiguous when there is a tie. The following statements make the possible quantifiers explicit.

Definition 2.1 (Every-minimizer form). For every finite graph $G = (V, E)$, every weight vector $(p_e)_{e \in E} \in [0, 1]^E$, every nonempty $A \subseteq V$, every $o, b \in V$, and every $a \in A$,

$$a \in A, \quad \rho_a \leq \rho_x \text{ for every } x \in A \implies L_a \leq R. \quad (5)$$

Definition 2.2 (Strict-minimizer form). For every finite graph $G = (V, E)$, every weight vector $(p_e)_{e \in E} \in [0, 1]^E$, every nonempty $A \subseteq V$, every $o, b \in V$, and every $a \in A$,

$$a \in A, \quad \rho_a < \rho_x \text{ for every } x \in A \setminus \{a\} \implies L_a \leq R. \quad (6)$$

Definition 2.3 (Positive-avoidance form). With the same closed quantifiers as in Definition 2.1, the implication (5) is required under the additional hypothesis $\mathbb{P}(D) > 0$.

Definition 2.4 (Existential form). For every finite graph $G = (V, E)$, every weight vector in $[0, 1]^E$, every nonempty $A \subseteq V$, and every $o, b \in V$, there is at least one $a \in \arg \min_{x \in A} \rho_x$ for which $L_a \leq R$.

Formulation	Status established in this note
Every minimizer, probabilities in $[0, 1]$	False; Theorem 3.2
Strict or unique minimizer	Open
Every minimizer with $\mathbb{P}(D) > 0$	Open; equivalent to the strict form
Some minimizing relay is good	Open in general
Minimizer of L_x , rather than ρ_x	True by strong Conjecture 2

Thus the unqualified assertion “Question 8 is proved” is not justified. The every-minimizer form is refuted, and the other natural readings still require a new argument.

3 Exact four-vertex calculation

The counterexample is most transparent as a symbolic three-parameter calculation.

Lemma 3.1 (Weighted four-vertex path). *Let*

$$V = \{o, a_1, a_2, b\}, \quad A = \{a_1, a_2\},$$

let $s, q, t \in [0, 1]$, and consider the path

$$o - a_1 - a_2 - b.$$

Give its three consecutive edges probabilities s, q, t , respectively, and give every other edge weight zero. Then

$$\mathbb{P}(U) = s, \quad R = sqt, \quad (7)$$

$$L_{a_1} = sqt, \quad L_{a_2} = st, \quad (8)$$

$$\rho_{a_1} = (1 - s)qt, \quad \rho_{a_2} = (1 - s)t. \quad (9)$$

Proof. Write X, Y, Z for the indicators that the three consecutive edges are open. The vertex o has only the first edge available, and $a_1 \in A$, so $U = \{X = 1\}$. The event $\{o \leftrightarrow b\}$ is $\{X = Y = Z = 1\}$. The connection $a_1 \leftrightarrow b$ requires $Y = Z = 1$, whereas $a_2 \leftrightarrow b$ requires only $Z = 1$. Independence now gives all six identities. $\square$

Theorem 3.2 (The every-minimizer form is false). *Definition 2.1 is false in the percolation model of the paper. The failure occurs with four distinct vertices and with $o, b \notin A$.*

Proof. In Lemma 3.1, set

$$s = 1, \quad q = \frac{1}{2}, \quad t = 1.$$

These weights are admissible because the model permits probabilities in $[0, 1]$. Equation (9) gives $\rho_{a_1} = \rho_{a_2} = 0$, so both relays are minimizers in (4). If the selected minimizer is a_2 , however,

$$L_{a_2} = 1 > \frac{1}{2} = R$$

by (7)–(8). This is precisely a failure of display (41). $\square$

Remark 3.3 (Why the obvious perturbation is not a strict counterexample). Keep $0 < s < 1$, $0 \leq q < 1$, and $t > 0$. Then $\rho_{a_1} < \rho_{a_2}$, so a_1 is the unique Question 8 minimizer; but $L_{a_1} = R = sqt$. The displayed boundary counterexample therefore does not refute Definition 2.2.

4 Diagnosis of the earlier argument

The already proved strong form of Conjecture 2 is

$$R \geq \min_{x \in A} L_x. \tag{10}$$

Consequently, a relay minimizing $L_x = \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)$ satisfies display (41). The earlier manuscript used exactly this observation, but it transcribed the selection rule in Question 8 as minimization of L_x . The printed selection rule is minimization of ρ_x .

Identity (3) says

$$\rho_x = \mathbb{P}(x \leftrightarrow b) - L_x.$$

It supplies no equivalence between the two rankings. In the counterexample, both ρ -scores vanish while $L_{a_1} = 1/2$ and $L_{a_2} = 1$. Hence the strong Conjecture 2 argument proves a useful neighboring statement, but it proves neither the strict nor the every-minimizer form of the printed Question 8.

5 Elementary cases that are settled

Proposition 5.1 (One relay). *If $A = \{a\}$, then $L_a = R$.*

Proof. On $U = \{o \leftrightarrow a\}$, the open clusters of o and a coincide. Thus $\mathbf{1}_{\{a \leftrightarrow b\}} = \mathbf{1}_{\{o \leftrightarrow b\}}$ on U . $\square$

Proposition 5.2 (The degenerate locus). *If $\mathbb{P}(D) = 0$, then every $x \in A$ minimizes ρ_x . Moreover, (10) guarantees at least one minimizing relay x with $L_x \leq R$, but Theorem 3.2 shows that another minimizing relay can fail the inequality.*

Proof. The hypothesis makes $\rho_x = \mathbb{P}(D, x \leftrightarrow b) = 0$ for every $x \in A$. The existence assertion is (10); the failure of the universal assertion was computed above. $\square$

Proposition 5.3 (Overlap with Question 7). *Suppose a is a Question 8 minimizer and also satisfies*

$$\mathbb{P}(a \leftrightarrow b) \leq \mathbb{P}(x \leftrightarrow b) \quad (x \in A).$$

Then $L_a \leq R$.

Proof. This is exactly the already machine-checked affirmative answer to Question 7, applied to the specified unconditional minimizer. $\square$

Proposition 5.4 (Observer as target). *If $b = o$, then every $a \in A$ satisfies $L_a \leq R$.*

Proof. Here $R = \mathbb{P}(U)$, whereas $L_a = \mathbb{P}(o \leftrightarrow A, a \leftrightarrow o) \leq \mathbb{P}(U)$. $\square$

6 Zero score under positive avoidance

The next lemma removes one boundary case from the open problem. It is stated without assuming that all edge weights lie strictly between zero and one.

Lemma 6.1 (Support-graph lemma). *Assume $\mathbb{P}(D) > 0$, let $a \in A$, and suppose $\rho_a = 0$. Then, up to a null event,*

$$\{a \leftrightarrow b\} \subseteq \{o \leftrightarrow b, o \leftrightarrow A\}. \quad (11)$$

In particular, $L_a \leq R$.

Proof. Let $E_1 = \{e : p_e = 1\}$ be the forced-open edges and $E_* = \{e : 0 < p_e < 1\}$ the genuinely random edges. Every prescribed subset of E_* , together with all of E_1 , has positive probability. Since $\mathbb{P}(D) > 0$, the component of o in the forced-open graph (V, E_1) contains no point of A .

Suppose that a supported configuration contains an open path from a to b , but does not connect o to b . Choose a simple open a -to- b path P . Form a new supported configuration by retaining all forced-open edges and retaining among the genuinely random edges only those belonging to P . It still connects a to b .

It cannot connect o to A . Indeed, a connection using no edge of $P \cap E_*$ would put a point of A in the forced-open component of o . A connection using such an edge would connect o to the path P , and hence to b , already in the original configuration. Both alternatives contradict the preceding paragraph or the choice of the original configuration. The new configuration therefore belongs to $\{a \leftrightarrow b\} \cap D$ and has positive probability, contradicting $\rho_a = 0$.

Thus $a \leftrightarrow b$ implies $o \leftrightarrow b$ almost surely. Whenever both connections hold, $o \leftrightarrow a$ through b ; since $a \in A$, the event U also holds. This proves (11), and taking probabilities gives $L_a \leq R$. $\square$

7 A pendant-relay reduction

The strict-minimizer version and the positive-avoidance every-minimizer version are essentially the same conjecture. This reduces the remaining work to a single precise target.

Lemma 7.1 (Pendant tie breaker). *Assume $\mathbb{P}(D) > 0$. Let $a \in A$ minimize ρ_x , and suppose $\rho_a > 0$. For $0 < q < 1$, add a new vertex $\hat{a}$, joined only to a by an edge of probability q , and replace a in the relay set:*

$$\hat{A} = (A \setminus \{a\}) \cup \{\hat{a}\}.$$

Use hats for the quantities (2) in the enlarged graph. Then $\hat{a}$ is the unique minimizer of the new Question 8 score. More precisely,

$$\hat{\rho}_{\hat{a}}(q) = q\rho_a, \quad (12)$$

$$\hat{\rho}_x(q) = q\rho_x + (1-q)\mathbb{P}(x \leftrightarrow b, o \leftrightarrow A \setminus \{a\}) \geq \rho_x \quad (x \in A \setminus \{a\}), \quad (13)$$

$$\hat{L}_{\hat{a}}(q) = qL_a, \quad (14)$$

$$\hat{R}(q) = qR + (1-q)\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A \setminus \{a\}). \quad (15)$$

Proof. Condition on the state of the new pendant edge. If it is open, wiring $\hat{a}$ to a identifies the new events with the original ones. If it is closed, $\hat{a}$ cannot connect to the old target b , and the new observer-to-relay event is $\{o \leftrightarrow A \setminus \{a\}\}$. This proves all four identities. Since

$$\{o \leftrightarrow A\} \subseteq \{o \leftrightarrow A \setminus \{a\}\},$$

the last probability in (13) is at least ρ_x , which proves the displayed lower bound. Therefore, for every old relay $x \neq a$,

$$\hat{\rho}_{\hat{a}} = q\rho_a < \rho_a \leq \rho_x \leq \hat{\rho}_x.$$

Thus $\hat{a}$ is the unique minimizer. $\square$

Theorem 7.2 (Equivalence of the two open cores). *The following global assertions are equivalent.*

- (i) *The strict-minimizer form, Definition 2.2, holds for every finite weighted graph.*
- (ii) *The positive-avoidance form, Definition 2.3, holds for every finite weighted graph and every choice among tied minimizers.*

Proof. Assume (i), suppose $\mathbb{P}(D) > 0$, and choose any minimizing relay a . If $\rho_a = 0$, Lemma 6.1 proves $L_a \leq R$. If $\rho_a > 0$, apply (i) to the unique minimizer $\hat{a}$ constructed in Lemma 7.1. Equations (14) and (15) give

$$qL_a \leq qR + (1-q)\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A \setminus \{a\}) \quad (0 < q < 1).$$

Letting $q \uparrow 1$ gives $L_a \leq R$. No limit theorem is needed for a formal proof: if $\Delta = L_a - R > 0$, then $0 < \Delta \leq 1$, and the choice $q = 1 - \Delta/4$ makes $q\Delta > (1-q)\mathbb{P}(o \leftrightarrow b, o \leftrightarrow A \setminus \{a\})$, contradicting the last display. Hence (ii) holds.

Conversely, suppose (ii) and let a be a strict minimizer. If $\mathbb{P}(D) > 0$, apply (ii). If $\mathbb{P}(D) = 0$, all the scores are zero, so a strict minimizer can exist only when $A = \{a\}$; Proposition 5.1 then applies. Thus (i) holds. $\square$

Remark 7.3. Theorem 7.2 does not restore the false every-minimizer claim on $\mathbb{P}(D) = 0$. The positive-avoidance hypothesis is indispensable in the stated equivalence.

8 Reduction to interior edge weights

Proposition 8.1 (Boundary closure for a strict minimizer). *It is enough to prove Definition 2.2 when every edge weight lies in $(0, 1)$.*

Proof. The claim is immediate for a singleton relay set. Otherwise, a strict minimizer has a positive gap

$$\delta := \min_{x \in A \setminus \{a\}} (\rho_x - \rho_a) > 0.$$

Every event probability on a finite graph is a multilinear polynomial in the edge weights. Approximate an arbitrary weight vector $p \in [0, 1]^E$ by vectors $p^{(n)} \in (0, 1)^E$. By continuity, the same relay a remains the strict minimizer for all sufficiently large n . The desired inequality at $p^{(n)}$, followed by another use of continuity, gives the inequality at p . $\square$

Combining Proposition 8.1, Theorem 7.2, and Lemma 6.1, the genuinely open case may be stated with all of the following assumptions:

$$p_e \in (0, 1), \quad |A| \geq 2, \quad \mathbb{P}(D) > 0, \quad \rho_a > 0, \quad \rho_a < \rho_x \ (x \neq a). \quad (16)$$

The conclusion still needed is $L_a \leq R$.

9 An exact event reduction and a proved sufficient condition

Fix $a \in A$, put $T = A \setminus \{a\}$, and define

$$E_a := \{o \leftrightarrow a, o \leftrightarrow T\}. \quad (17)$$

Lemma 9.1 (Cancellation on the connected branch). *For every $a \in A$,*

$$L_a - R = \mathbb{P}(a \leftrightarrow b, E_a) - \mathbb{P}(o \leftrightarrow b, E_a). \quad (18)$$

Proof. Partition U according as $o \leftrightarrow a$ or $o \nleftrightarrow a$. On $\{o \leftrightarrow a\}$, the clusters of o and a coincide, so the two contributions cancel. On $\{o \nleftrightarrow a\}$, the event U is exactly $\{o \leftrightarrow T\}$, giving (18). $\square$

The next proposition records a route already supported by the avoided first-relay machinery. It is a sufficient condition, not a proof that the Question 8 selection rule implies the condition.

Let

$$T_+ := \{x \in T : \mathbb{P}(x \leftrightarrow a) > 0\}. \quad (19)$$

For a relay $x \in T_+$, set

$$m_x^a := \mathbb{P}(x \leftrightarrow b \mid x \leftrightarrow a). \quad (20)$$

Choose an injective rank on T_+ that is nondecreasing in these conditional probabilities, breaking ties arbitrarily. Let J_x be the event that $o \leftrightarrow a$ and x is the first relay of T_+ , in that rank, reached by o . These events are disjoint and their union is E_a , up to a null event. Indeed, if $x \in T \setminus T_+$, then $\{o \leftrightarrow a, o \leftrightarrow x\} \subseteq \{x \leftrightarrow a\}$ is null. A finite union over the inactive relays is still null, so deleting them makes no contribution to E_a .

Proposition 9.2 (A formalizable sufficient criterion). *Assume $\mathbb{P}(o \leftrightarrow a) > 0$. If*

$$\mathbb{P}(a \leftrightarrow b, a \leftrightarrow o) \mathbb{P}(E_a) \leq \mathbb{P}(a \leftrightarrow o) \sum_{x \in T_+} \mathbb{P}(J_x) m_x^a, \quad (21)$$

then $L_a \leq R$. In particular, it is sufficient that

$$\mathbb{P}(a \leftrightarrow b \mid a \leftrightarrow o) \leq \min_{x \in T_+} \mathbb{P}(x \leftrightarrow b \mid x \leftrightarrow a), \quad (22)$$

provided E_a has positive probability.

Proof. The assumption $\mathbb{P}(o \leftrightarrow a) > 0$ in particular implies $a \neq o$. The denominator-free form of the two-cluster BHK inequality, conditioned on $o \leftrightarrow a$, gives

$$\mathbb{P}(o \leftrightarrow a)\mathbb{P}(a \leftrightarrow b, E_a) \leq \mathbb{P}(a \leftrightarrow b, a \leftrightarrow o)\mathbb{P}(E_a). \quad (23)$$

The avoided first-relay inequality with avoided set $\{a\}$, active relay set T_+ , and cluster functional $K \mapsto \mathbf{1}_{\{b \in K\}}$ gives

$$\mathbb{P}(o \leftrightarrow b, E_a) \geq \sum_{x \in T_+} \mathbb{P}(J_x) m_x^a. \quad (24)$$

Combining (21), (23), and (24), then cancelling the positive factor $\mathbb{P}(o \leftrightarrow a)$ and using Lemma 9.1, proves the first claim. For (22), use $\sum_{x \in T_+} \mathbb{P}(J_x) = \mathbb{P}(E_a)$. If E_a has positive probability, then T_+ is nonempty, so the minimum in (22) is defined. $\square$

The missing bridge is now explicit: no proof is presently supplied that the Question 8 hypothesis $\rho_a \leq \rho_x$ forces (21). Numerical agreement of the final inequality cannot replace this implication.

9.1 A sharper conjectural bridge

The following inequality is a useful research target, but it is *not proved in this note*. Work under the strict assumptions (16). For $x \in T$, put

$$d_x := \mathbb{P}(x \leftrightarrow a), \quad \beta_x := \frac{\rho_x - \rho_a}{d_x}. \quad (25)$$

Every d_x is positive: if $x \leftrightarrow a$ almost surely, then the indicators of $x \leftrightarrow b$ and $a \leftrightarrow b$ agree almost surely, forcing $\rho_x = \rho_a$ and contradicting strictness. Rank T increasingly by β_x , and define the associated first-relay events J_x on $\{o \leftrightarrow a\}$ as above. The candidate inequality is

$$\boxed{R - L_a \geq \sum_{x \in T} \mathbb{P}(J_x) \frac{\rho_x - \rho_a}{\mathbb{P}(x \leftrightarrow a)}}. \quad (26)$$

It would immediately settle the strict case because every term on the right is nonnegative. It has survived small-graph computational tests, but that is evidence only.

The direct attempted derivation has a specific obstruction. The natural cluster projection whose avoided relay integral equals $\rho_x - \rho_a$ is not monotone, so the existing avoided first-relay theorem cannot be applied to it. Any proof of (26) therefore needs either a monotone correction whose error has a controlled sign or a new signed version of the first-relay inequality. A Lean development may state (26) as a closed proposition for future work, but must not import it as an axiom or use it in the proved chain above.

10 Lean handoff

The formalization should keep false, proved, and open declarations in separate namespaces or files. A minimal declaration list is as follows.

Q8.1 U, q8L, q8Score, and q8R, corresponding to (1)–(2).

Q8.2 IsQ8Min and IsQ8StrictMin.

Q8.3 Closed propositions `Question8EveryMin`, `Question8Strict`, `Question8Positive`, and `Question8ExistsGoodMin`. The first is to be refuted; the other three are statements only.

Q8.4 `q8_total_split`, the identity (3).

Q8.5 The six identities (7)–(9) on `Fin 4`, followed by `not_question8EveryMin`.

Q8.6 `q8_singleton`, `q8_bEq0`, and `q8_degenerate_allMin`.

Q8.7 A support-atom lemma: a configuration has positive product-Bernoulli mass exactly when it contains every pair of weight one and contains no pair of weight zero. Use it in `q8_zeroScore`, Lemma 6.1, rather than introducing conditional probabilities.

Q8.8 The four pendant identities (12)–(15), then `q8Strict_iff_q8Positive`. Concretely, enlarge `Fin n` to `Fin (n+1)`, embed old vertices with `Fin.castSucc`, use `Fin.last n` for $\hat{a}$, and define the new `Sym2` weight by cases: the pendant pair has weight q , embedded old pairs retain w , and all other new pairs have weight zero.

Q8.9 `q8_target_split`, identity (18).

Q8.10 `q8_of_firstRelayCriterion`, Proposition 9.2, importing the already formalized avoided first-relay theorems `KNA11.genY_all` and `sum_le_setIntegral_of_genY`, together with `BHK2006_twoClusterConditionalAssociation.negCorrelation`.

Q8.11 A statement-only declaration `Question8Bridge` for (26). It is not a theorem and must have no role in the proofs of the preceding declarations.

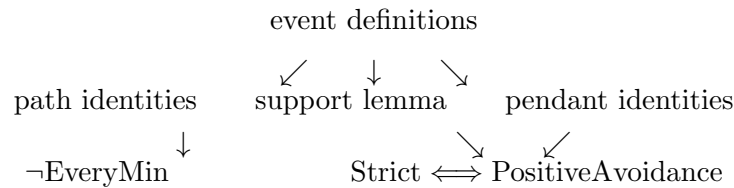
For Proposition 8.1, reuse `prodBernoulli_real_continuous`; the positive strict-minimum gap keeps the same relay minimizing along the approximating sequence.

No theorem named `question8_holds` should be introduced unless the open core (16) is actually proved. In particular, the theorem

$$R \geq \min_{x \in A} L_x$$

must not be used with a ρ -minimizer without an additional argument relating the two rankings.

Suggested dependency graph



Separately,

$$\text{BHK} + \text{avoided first-relay} + (18) \implies \text{Proposition 9.2.}$$

This organization lets a checker certify every result in the present note without inserting a placeholder proof for the genuinely open implication.

11 Final status

The exact conclusion is:

1. The printed score is $\rho_x = \mathbb{P}(x \leftrightarrow b, o \leftrightarrow A)$.
2. The every-minimizer reading over weights in $[0, 1]$ is false by an exact four-vertex counterexample.
3. The previous manuscript proved the different statement obtained by minimizing $L_x = \mathbb{P}(o \leftrightarrow A, x \leftrightarrow b)$.
4. The strict-minimizer version and the every-minimizer version under $\mathbb{P}(o \leftrightarrow A) > 0$ are equivalent, but no complete proof of either is given here.
5. The remaining proof obligation is the implication from the strict ordering of the ρ_x 's in (16) to $L_a \leq R$, equivalently the inequality in Lemma 9.1.

References

- [1] G. Kozma and S. Nitzan, *A reduction of the $\theta(p_c) = 0$ problem to a conjectured inequality*, arXiv:2401.12397, 2024. Question 8 appears on p. 36.